BAN 100
ASSIGNMENT 2: ANOVA

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Problem 1:

HYPOTHESIS:
H0: Ownership of stocks is not affected by age.
Ha: Ownership of stocks is affected by at least one of the age groups.

CONCLUSION:
P value is less than α (0.0405<0.05).
We reject the null hypothesis (Ho).

Therefore, ownership of stocks is affected by the age groups.

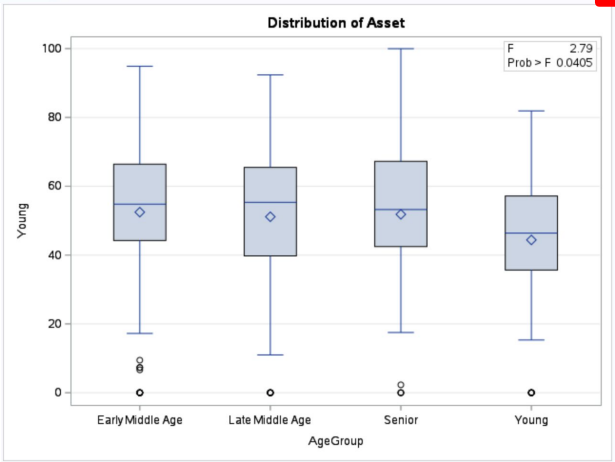
The ANOVA Procedure

Dependent Variable: Asset Young

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	3741.3636	1247.1212	2.79	0.0405
Error	362	161870.9817	447.1574		
Corrected Total	365	165612.3453			

R-Square	Coeff Var	Root MSE	Asset Mean
0.022591	42.14046	21.14610	50.18003

Source	DF	Anova SS	Mean Square	F Value	Pr > F
AgeGroup	3	3741.363610	1247.121203	2.79	0.0405



Problem 2 - A:

HYPOTHESIS - A:
H0: There is interaction between gender and education in holding jobs.
Ha: There is no interaction between gender and education in holding jobs.

CONCLUSION:
P value is greater than α (0.8915>0.05).
We fail to reject the null hypothesis (Ho).

Therefore, There is interaction between gender and education in holding jobs.

The ANOVA Procedure

Dependent Variable: Number_Jobs Female_E1

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	153.3500000	21.9071429	2.17	0.0467
Error	72	726.2000000	10.0861111		
Corrected Total	79	879.5500000			

R-Square	Coeff Var	Root MSE	Number_Jobs Mean
0.174351	30.46392	3.175864	10.42500

Source	DF	Anova SS	Mean Square	F Value	Pr > F
Ed_level	3	135.8500000	45.2833333	4.49	0.0060
Gender	1	11.2500000	11.2500000	1.12	0.2944
Ed_level*Gender	3	6.2500000	2.0833333	0.21	0.8915

Problem 2 - B:

HYPOTHESIS - B:
H0: There are differences in holding jobs between men and women.
Ha: There are no differences in holding jobs between men and women.

CONCLUSION:
P value is greater than α (0.2944>0.05). We fail to reject the null hypothesis (Ho).

Therefore, There are differences in holding jobs between men and women.

The ANOVA Procedure

Dependent Variable: Number_Jobs Female_E1

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	153.3500000	21.9071429	2.17	0.0467
Error	72	726.2000000	10.0861111		
Corrected Total	79	879.5500000			

R-Square	Coeff Var	Root MSE	Number_Jobs Mean
0.174351	30.46392	3.175864	10.42500

Source	DF	Anova SS	Mean Square	F Value	Pr > F
Ed_level	3	135.8500000	45.2833333	4.49	0.0060
Gender	1	11.2500000	11.2500000	1.12	0.2944
Ed_level*Gender	3	6.2500000	2.0833333	0.21	0.8915

Problem 2 - C:

HYPOTHESIS - C:
H0: There are differences in holding jobs between the educational levels.
Ha: There are no differences in holding jobs between the educational levels.

CONCLUSION:
P value is less than α (0.0060<0.05).
We reject the null hypothesis (Ho).

Therefore, There are no differences in holding jobs between the educational levels.

The ANOVA Procedure

Dependent Variable: Number_Jobs Female_E1

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	153.3500000	21.9071429	2.17	0.0467
Error	72	726.2000000	10.0861111		
Corrected Total	79	879.5500000			

R-Square	Coeff Var	Root MSE	Number_Jobs Mean
0.174351	30.46392	3.175864	10.42500

Source	DF	Anova SS	Mean Square	F Value	Pr > F
Ed_level	3	135.8500000	45.2833333	4.49	0.0060
Gender	1	11.2500000	11.2500000	1.12	0.2944
Ed_level*Gender	3	6.2500000	2.0833333	0.21	0.8915